

Information Design Perspective on Calibration

PART I – INTRODUCTION

YIDING FENG, HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

JOINT TUTORIAL WITH WEI TANG, CUHK

EC 2025

Calibration 101: What is **calibration**?

Example: Bitcoin Up or Down?



Question: How to measure the **quality** of **prediction**?

Example: Bitcoin Up or Down?

Predictions in past 100 days:

	day 1	day 2	day 3	day 4	...	day 99	day 100
Prediction (prob for Up)	50%	20%	20%	50%		70%	20%
Outcome	Up	Up	Down	Down		Down	Up

Natural criterion of **Good** predictions:

➤ “among all days with prediction = X%, X% of those days are **UP**”

Calibrated Predictor

- Binary random outcome $Y \in \{0, 1\}$ “Down vs. Up”
- (Possibly random) predictor $F \in \Delta([0, 1])$ “ $p \in [0, 1]$: chance for Up”

Definition [Dawid, JASA’82][Foster Vohra, Biometrika’98].

Predictor F is **calibrated** if for **every** prediction $q \in [0, 1]$

$$\mathbb{E}[Y|F = q] = q$$

“condition on prediction, true Up probability is equal to prediction”

(Mis-)Calibrated Predictors

Example. Suppose $Y \sim \text{Bern}(0.5)$.

calibrated



Predict base rate $\mathbb{E}[Y]$

$$F \equiv \mathbb{E}[Y] = 0.5$$



Predict 100% (resp. 0%) if **Up** (resp. **Down**)

$$F \equiv \mathbb{I}\{Y = 1\}$$

mis-calibrated



Predict 100% and 0% uniformly

$$F \sim \text{Bern}(0.5)$$

F independent of Y

$$\mathbb{E}[Y|F = 1] = 0.5$$

$$\mathbb{E}[Y|F = 0] = 0.5$$



Predict 100% (resp. 0%) if **Down** (resp. **Up**)

$$F \equiv \mathbb{I}\{Y = 0\}$$

$$\mathbb{E}[Y|F = 1] = 0$$

$$\mathbb{E}[Y|F = 0] = 1$$

Calibration Error

Definition

The expected calibration error (ECE) of predictor F is

$$\text{ECE}[F] \stackrel{\text{def}}{=} \mathbb{E}_{q \sim F}[|q - \mathbb{E}[Y|q]|]$$

We say a predictor F is ε -calibrated if $\text{ECE}[F] \leq \varepsilon$.

Example (cont.) Suppose $Y \sim \text{Bern}(0.5)$. Compute ECE:



Predict base rate $\mathbb{E}[Y]$

$$F \equiv \mathbb{E}[Y] = 0.5$$

$$\text{ECE}[F] = |0.5 - 0.5| = 0$$



Predict 100% (resp. 0%) if **Up** (resp. **Down**)

$$F \equiv \mathbb{I}\{Y = 1\}$$

$$\text{ECE}[F] = |1 - 1| \cdot 0.5 + |0 - 0| \cdot 0.5 = 0$$



Predict 100% and 0% uniformly

$$F \sim \text{Bern}(0.5)$$

F independent of Y

$$\text{ECE}[F] = |0.5 - 1| \cdot 0.5 + |0.5 - 0| \cdot 0.5 = 0.5$$



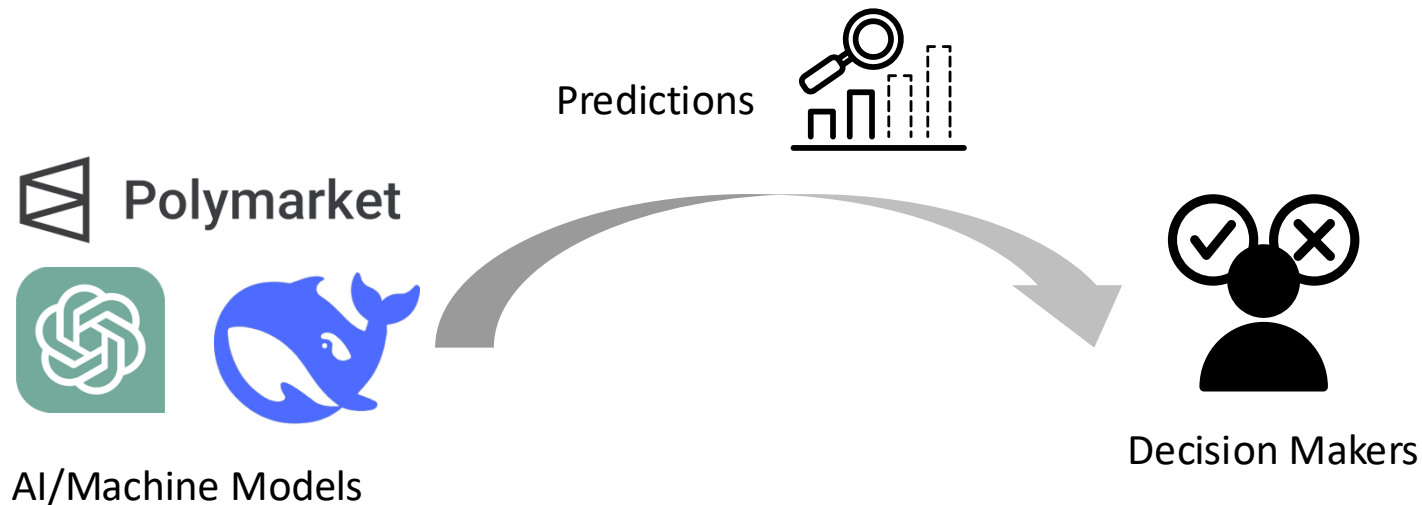
Predict 100% (resp. 0%) if **Down** (resp. **Up**)

$$F \equiv \mathbb{I}\{Y = 0\}$$

$$\text{ECE}[F] = |0 - 1| \cdot 0.5 + |1 - 0| \cdot 0.5 = 1$$

Why calibration?

Machine-in-the-loop Decision Making



Other applications of machine predictors:

- Digital advertising
- Criminal Justice
- Healthcare
- ...

Machine Predictions in High-Stakes Decision Making



- **Digital Advertising Auction: Google's CTR models**
 - offer CTR prediction for advertisers to guide their auction bidding



- **Criminal Justice: COMPAS (Northpointe/Equivant)**
 - offer recidivism predictions to guide judges on bail decisions



- **Healthcare: Epic's Deterioration Index / Sepsis Model**
 - provides disease predictions to guide doctor's clinic decision

Trustworthiness in Machine Learning

“**calibrated** predictor $\Rightarrow$ **reliable** to be used to make decisions”

Downstream decision maker (agent)

- Unobserved binary random outcome $Y \in \{0, 1\}$ “**Up** vs. **Down**”
- Decides action $a \sim \mathcal{A}$
- Receives agent utility $u(a, Y) \in [0, 1]$ “**Long** vs **Short** Bitcoin?”

Informal Theorem [Kleinberg Leme Schneider Teng, COLT’23].

Given any predictor F , suppose agent **naively best respond** to prediction $q \sim F$, i.e.,

$$a^* = \max_{a \in \mathcal{A}} \mathbb{E}_{Y \sim \text{Bern}(q)}[u(a, Y)]$$

then agent’s **regret** is at most $\text{ECE}[F]$.

Regret = Payoff by best-responding to $\mathbb{E}[Y \mid q]$ — Payoff by best-responding to q

Trustworthiness in Machine Learning

Takeaways

1. **No-regret**: Making decision based on perfectly/almost calibrated predictor ensures zero/small regret.
2. **No need** to know details of F

Related Work

Econ/Stats literature [Foster Vohra, GEB'97, Biometrika'98], [Hart Mas-Colell, ECMA'00], [Foster Hart, JPE'21], [Foster Hart, TE'23], [Guo Shmaya, TE'23] ...

CS/ML literature:

- **Calibration in neural network/LLMs** [Guo Pleiss Sun Weinberger, ICML'17], [Percy Liang et al., TMLR'23], [Kalai Vempala, STOC'24], ...
- **Online calibration** [Qiao Valiant, STOC'21], [Qiao Zhang, COLT'24], [Okoroafor Kleinberg Sun, AISTATS'24], [Dagan Daskalakis Fishelson Golowich Kleinberg Okoroafor, STOC'25], ...
- **Multi-calibration** [Hébert-Johnson Kim Reingold Rothblum, ICML'18], [Hu Peale, ITCS'23], [Casacuberta Dwork Vadhan, STOC'24], ...
- **Calibration in Decision-making** [Camara Hartline Johnsen FOCS'20], [Rothblum Yona, ITCS'23], [Kleinberg Leme Schneider Teng, COLT'23], [Roth Shi, EC'24], [Jain Vianney, EC'24], [Collina Roth Shao, EC'24], [Hu Wu, FOCS'24], [Feng Tang, SODA'26] ...

OR/MS literature [Elmachtoub Grigas, MS'22], [Jens Witkowski et al, MS'23], [Tsirtsis Tabibian Khajehnejad Singla Scholkopf Gomez-Rodriguez, MS'24] ...

Related Work

On calibration of modern neural networks

[C Guo](#), [G Pleiss](#), [Y Sun](#), [KQ Weinberger](#)

International conference on machine learning, 2017 • [proceedings.mlr.press](#)

Abstract

Confidence calibration—the problem of predicting probability estimates representative of the true correctness likelihood—is important for classification models in many applications. We discover that modern neural networks, unlike those from a decade ago, are poorly calibrated. Through extensive experiments, we observe that depth, width, weight decay, and Batch Normalization are important factors influencing calibration. We evaluate the performance of various post-processing calibration methods on state-of-the-art

SHOW MORE ▾

☆ Save  Cite Cited by 8375 Related articles All 8 versions 

Information Design 101: Bayesian Persuasion

Core idea: a *sender* commits to an information structure to influence a *receiver's* action

Basic model [Kamenica Gentzkow, AER'11]

- State $\omega \in \Omega$ drawn from prior μ
- Sender *commits* to **signaling scheme** $\pi: \Omega \rightarrow \Delta(\Sigma)$, where signal $\sigma \in \Sigma$ is revealed to receiver
- Receiver observes σ , updates belief via **Bayes' rule**, choose action $a \in \mathcal{A}$ to maximize their utility
- Sender chooses π to maximize their own utility, anticipating receiver's best response.

Key insight: Sender cannot lie after seeing ω , but can *design what is revealed* in advance.

Example: ad platform (sender) designs how to report predicted CTR to advertiser (receiver), who then decides how much to bid

Calibration: Information Design Perspective

Calibration

- Binary random outcome $Y \in \{0, 1\}$
-

- Feature/Context $X_i \sim D \in \Delta(\mathcal{X})$
-

- Each X_i has true prob. $p_i \in [0, 1]$

for outcome realization:

$$\mathbb{E}[Y|X = X_i] \stackrel{\text{def}}{=} p_i$$

- Predictor $F: \mathcal{X} \rightarrow \Delta([0, 1])$
-

Calibrated predictor requires that:

$$q = \mathbb{E}[Y|q] = \frac{\sum_{X_i} \mathbb{P}_{X \sim D}(X = X_i) \cdot F(q | X_i) \cdot p_i}{\sum_{X_i} \mathbb{P}_{X \sim D}(X = X_i) \cdot F(q | X_i)}, \quad \forall q \in \text{supp}(F)$$

Calibration: Information Design Perspective

Calibration

- Binary random outcome $Y \in \{0, 1\}$
- Feature/Context $X_i \sim D \in \Delta(\mathcal{X})$
- Each X_i has true prob. $p_i \in [0, 1]$
for outcome realization:

$$\mathbb{E}[Y|X = X_i] \stackrel{\text{def}}{=} p_i$$

- Predictor $F: \mathcal{X} \rightarrow \Delta([0, 1])$

Signaling scheme

- State space $\{p_i\}$
- Prior dist. is $\mathbb{P}(p = p_i) = \mathbb{P}_{X \sim D}(X = X_i)$

- Signal Space $\Sigma = [0, 1]$, signaling scheme
 $\pi: \{p_i\} \rightarrow \Delta(\Sigma)$

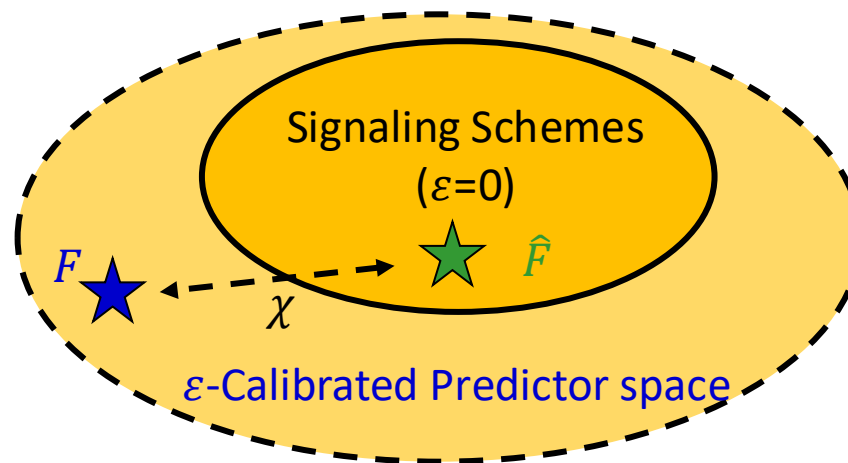
Calibrated predictor $\Leftrightarrow$ signaling scheme where signal is posterior mean

$$q = \mathbb{E}[Y|q] = \frac{\sum_{X_i} \mathbb{P}_{X \sim D}(X = X_i) \cdot F(q | X_i) \cdot p_i}{\sum_{X_i} \mathbb{P}_{X \sim D}(X = X_i) \cdot F(q | X_i)}, \quad \forall q \in \text{supp}(F)$$

Two-Step View for Predictors

Observation: generating ε -calibrated predictor F is equivalent to

- generating **(perfectly) calibrated predictor $\hat{F}$** (ECE $\varepsilon = 0$)
 - perfectly calibrated predictor** $\Leftrightarrow$ signaling scheme where **signals = posterior means**
- “miscalibrating” $\hat{F}$ into F with ε -calibration budget (denote this **miscalibration as χ**)



Examples via Two-Step Approach

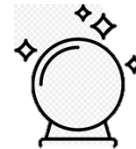
Define $\chi(q, q')$: frequency of miscalibrating true prob. q to prediction q'

calibrated predictor



Predict base rate $\mathbb{E}[Y]$

$$\hat{F} \equiv \mathbb{E}[Y] = 0.5$$



Predict 100% (0%) if **Up** (**Down**)

$$\hat{F} \equiv \mathbb{I}\{Y = 1\}$$

miscalibrated predictor



Predict 100% and 0% uniformly

$$F \sim \text{Bern}(0.5)$$
$$F \text{ independent of } Y$$



Predict 100% (0%) if **Down** (**Up**)

$$F \equiv \mathbb{I}\{Y = 0\}$$

Two-Step View for Predictors

Space of ϵ -calibrated predictor can be characterized as linear polytope:

Variable $\hat{F}$: perfectly calibrated predictor

Variable $\chi(q, q')$: frequency of miscalibrating **true prob.** q to **prediction** q'

Linear constraints:

$$\sum_{q \in [0,1]} \sum_{q' \in [0,1]} \chi(q, q') \cdot |q' - q| \leq \epsilon$$

χ satisfies ϵ -ECE budget

similar to budget constraint in auction design

$$\sum_{q' \in [0,1]} \chi(q, q') = \hat{F}(q), \forall q$$

χ is consistent with $\hat{F}$

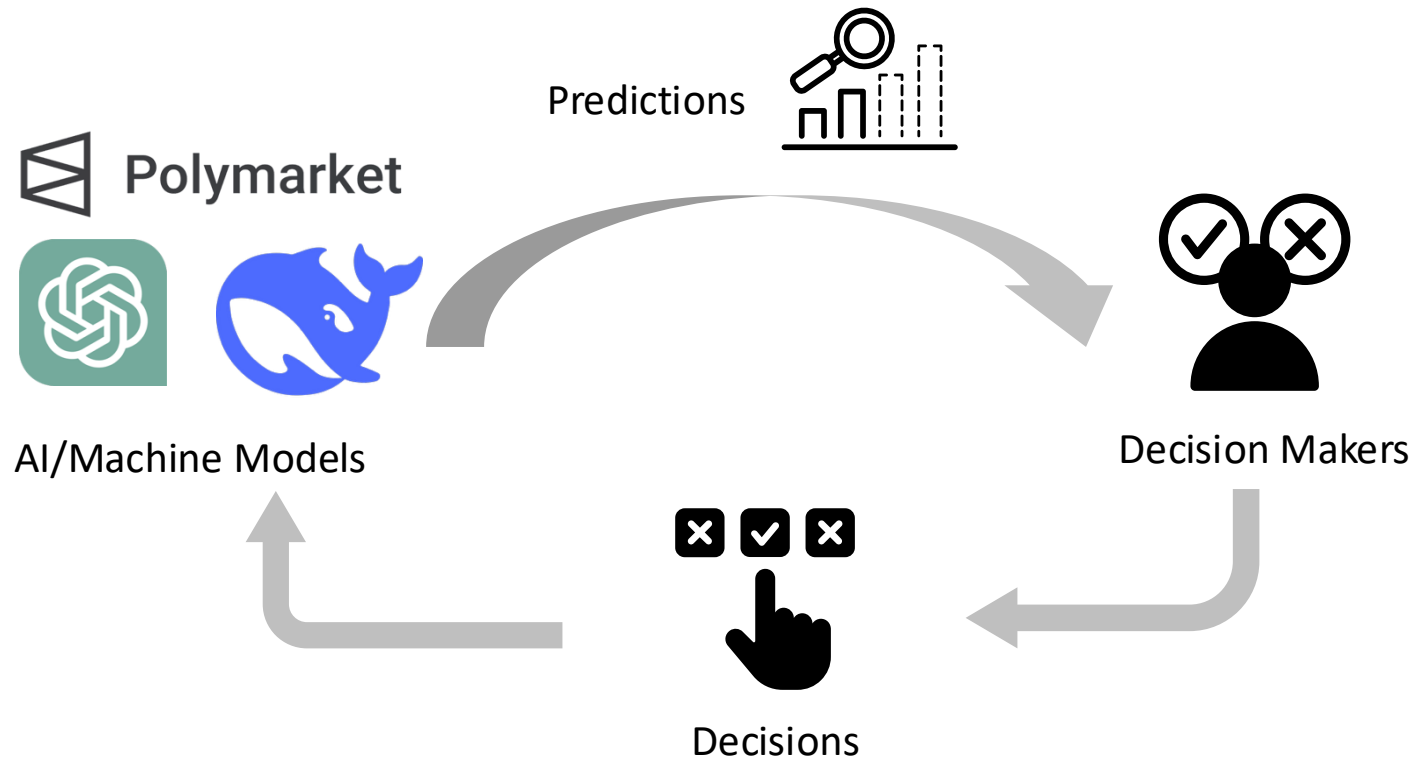
$\hat{F}$ is perfectly calibrated

Captured by **mean-preserving contractions (MPC) constraint**

i.e., 2nd-order stochastic dominance + identical mean
widely studied in information design literature

Takeaway: Useful tool for characterizing/computing (near)-optimal predictors²⁰

Application: Predictor Design under Incentive Misalignment



Decision maker's action may also affect AI designer's utility

Application: Predictor Design under Incentive Misalignment

[Informal Research Questions]

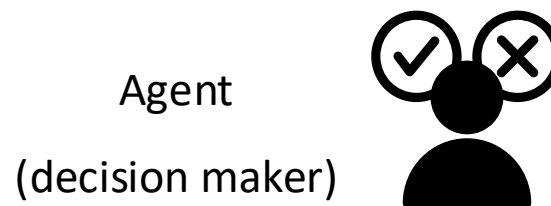
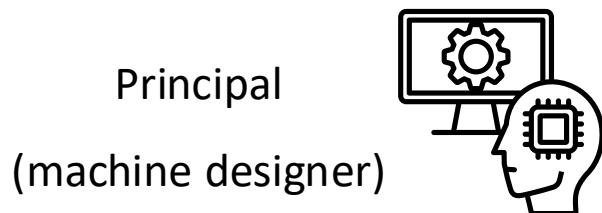
[Q1] Given a **calibration error budget**, what is the **optimal predictor**, especially when there exists **incentive misalignment** between the principal and the agent?

[Q2] Can we **compute** this optimal predictor or an approximately optimal predictor in **an efficient way**?

Persuasive Calibration

- Binary random outcome $Y \in \{0, 1\}$
- Feature/Context X sampled from feature dist. $D \in \Delta(\mathcal{X})$
- Each feature X has true prob. $p_X \in [0, 1]$ for outcome realization:

$$\mathbb{E}[Y|X] \stackrel{\text{def}}{=} p_X$$



- Knows D, p_X , but not Y
- Principal utility: $u^P(a)$

- Decides action $a \sim \mathcal{A}$
- Agent utility: $u^A(a, Y)$

Persuasive Calibration

random mapping from feature to predictions

Goal: identify principal's optimal predictor $F: \mathcal{X} \rightarrow \Delta([0,1])$ subject to ECE constraint

$$\max_{F: \mathcal{X} \rightarrow \Delta([0,1])} \quad u^P(F) = \mathbb{E}_{X \sim D} \mathbb{E}_{q \sim F(X)} [u^P(a^*(q))]$$

s. t.

$$\text{ECE}[F] \leq \varepsilon \quad \text{ECE constraint, } \varepsilon \text{ is pre-specified ECE budget}$$

$$a^*(q) = \underset{a \in \mathcal{A}}{\text{argmax}} \mathbb{E}_{Y \sim \text{Bern}(q)} [u^A(a, Y)], \quad \forall q \in \text{supp}(F)$$

agent **naively best responds**

trustworthiness ensured by ε -ECE above

Bayesian Persuasion	Persuasive Calibration
Knowing Priors & All signaling details	

Persuasive Calibration

random mapping from feature to predictions

Goal: identify principal's optimal predictor $F: \mathcal{X} \rightarrow \Delta([0,1])$ subject to ECE constraint

$$\begin{aligned}
 & \max_{F: \mathcal{X} \rightarrow \Delta([0,1])} && u^P(F) = \mathbb{E}_{X \sim D} \mathbb{E}_{q \sim F(X)} [u^P(a^*(q))] \\
 & \text{s. t.} && \text{ECE}[F] \leq \varepsilon \quad \text{ECE constraint, } \varepsilon \text{ is pre-specified ECE budget} \\
 & && a^*(q) = \underset{a \in \mathcal{A}}{\operatorname{argmax}} \mathbb{E}_{Y \sim \text{Bern}(q)} [u^A(a, Y)], \quad \forall q \in \text{supp}(F)
 \end{aligned}$$

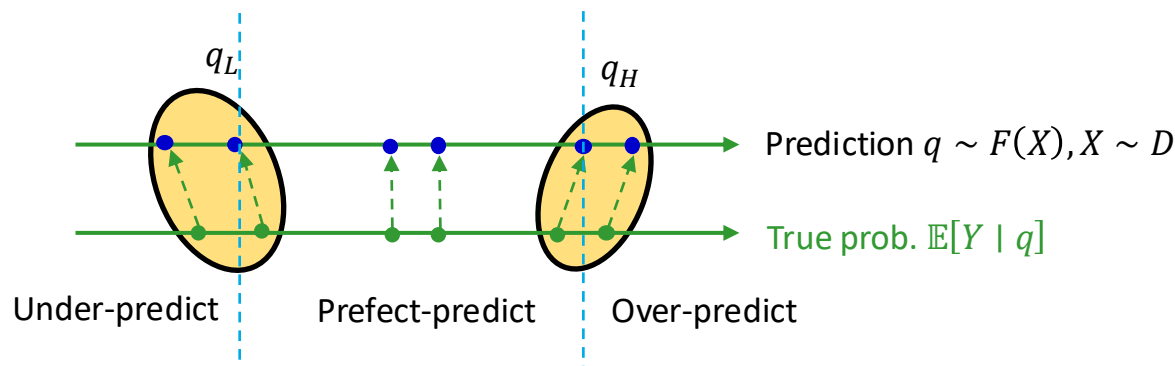
agent **naively best responds**
trustworthiness ensured by ε -ECE above

Bayesian Persuasion	Persuasive Calibration
Knowing Priors & All signaling details	
Bayesian Belief Update	
Commitment	Calibration

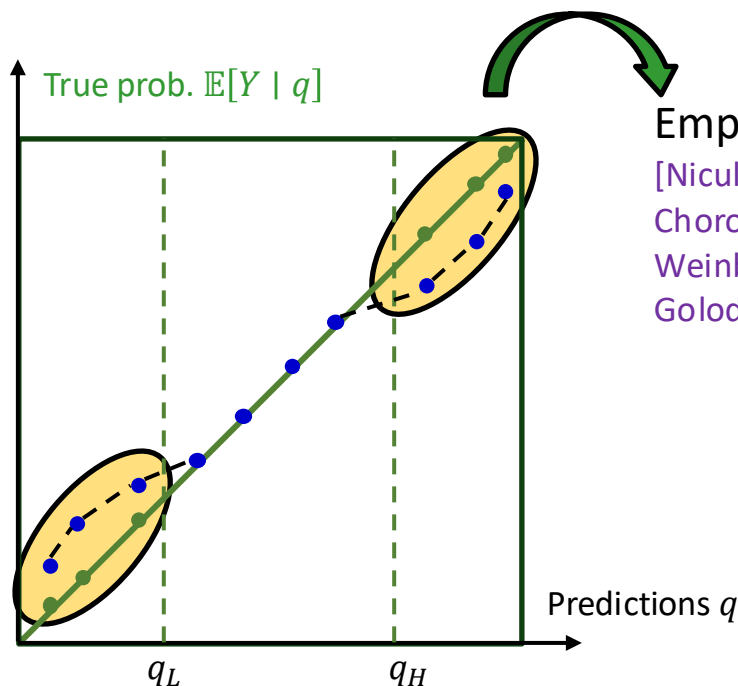
Characterizing Optimal ε -Calibrated Predictor

Theorem [Feng Tang, SODA'26]. In the optimal ε -calibrated predictor, there exists $0 \leq q_L \leq q_H \leq 1$ such that

- **[Miscalibration structure]** Predictions $q \geq q_H$ ($q \leq q_L$) over-predict (under-predict) true conditional probability.



Over-/under-confident Predictions in ML



Empirical evidence:

[Niculescu-Mizil Caruana, ICML'05] [Pereyra Tucker Chorowski Kaiser Hinton, ICLR'17] [Guo Pleiss Sun Weinberger, ICML'17] [Mukhoti Kulharia Sanyal Golodetz Torr Dokania, NeurIPS'20] ...

Under-predict Prefect-predict Over-predict

Takeaways

Regardless of loss functions, as long as there is calibration error, it must happen on extreme predictions.

Characterizing Optimal ε -Calibrated Predictor

Theorem [Feng Tang, SODA'26]. In the optimal ε -calibrated predictor, there exists $0 \leq q_L \leq q_H \leq 1$ such that

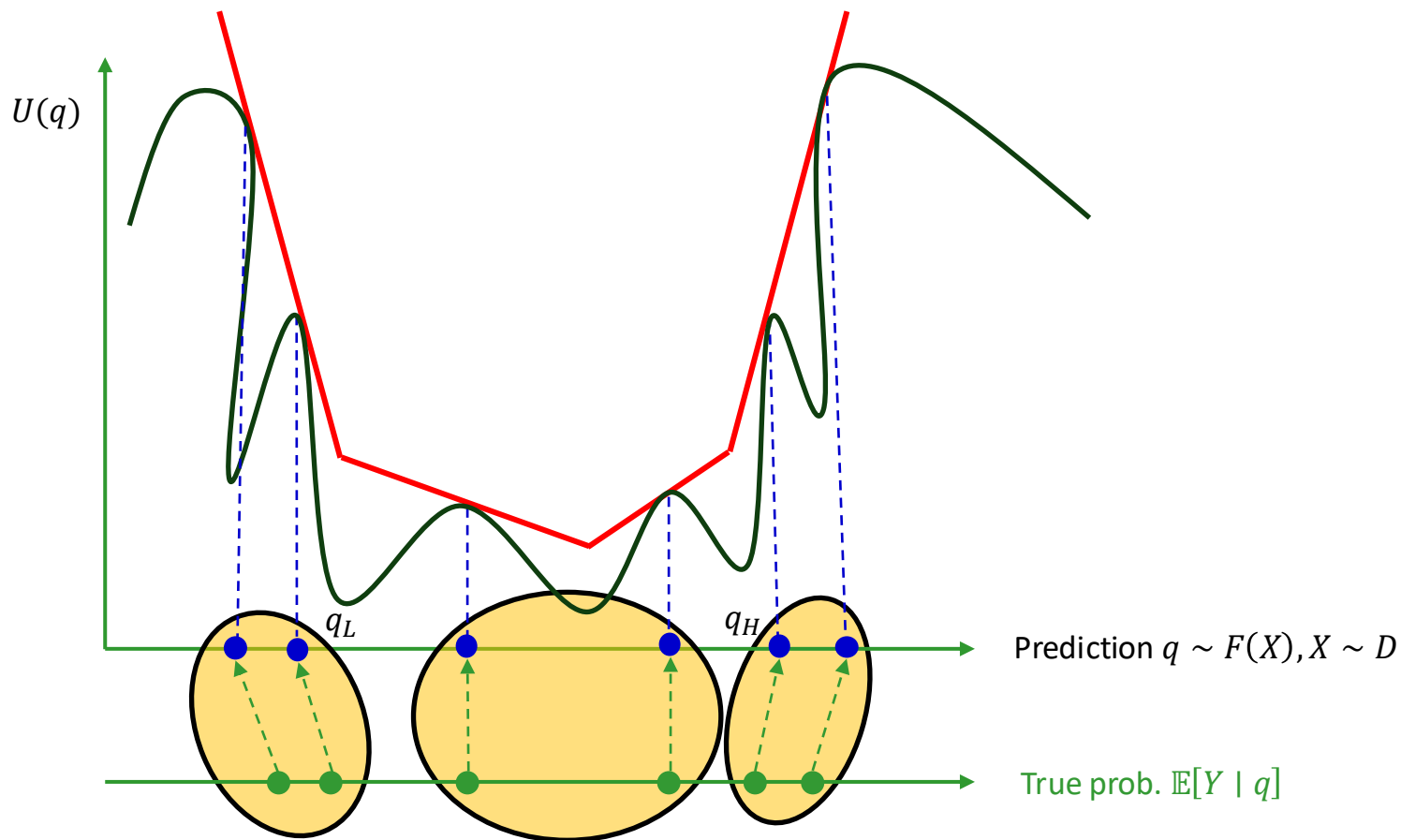
- [Miscalibration structure] Predictions $q \geq q_H$ ($q \leq q_L$) over-predict (under-predict) true conditional probability.
- [Payoff structure] For all predictions $q \in \text{supp}(F)$, the derivative of principal's indirect utility function $U'(q)$ is increasing in q , and satisfies

$$U'(q) = \alpha \text{ for } q \geq q_H; U'(q) = -\alpha \text{ for } q \leq q_L$$

$U'(q)$: “marginal utility gain by miscalibrating q ”

Indirect utility $U(q) \stackrel{\text{def}}{=} u^P(a^*(q))$ with $a^*(q) \stackrel{\text{def}}{=} \operatorname{argmax}_{a \in \mathcal{A}} \mathbb{E}_{Y \sim \text{Bern}(q)}[u^A(a, Y)]$

Payoff Structure of Optimal ε -Calibrated Predictor



Proof Sketch via Two-Step LP

Two-step view **linear program** for principal's problem:

Variable $\hat{F}$: perfectly calibrated predictor

Variable $\chi(q, q')$: frequency of miscalibrating **true prob.** q to **prediction** q'

$$\max_{\hat{F}, \chi} \quad U^P(\chi) := \sum_{q \in [0,1]} \sum_{q' \in [0,1]} \chi(q, q') \cdot U^P(q')$$

$$s. t. \quad \sum_{q \in [0,1]} \sum_{q' \in [0,1]} \chi(q, q') \cdot |q' - q| \leq \varepsilon \quad \chi \text{ satisfies } \varepsilon\text{-ECE budget}$$

similar to budget constraint in auction design

$$\sum_{q' \in [0,1]} \chi(q, q') = \hat{F}(q), \forall q$$

χ is consistent with $\hat{F}$

$\hat{F}$ is perfectly calibrated

Captured by **mean-preserving contractions (MPC) constraint**

i.e., 2nd-order stochastic dominance + identical mean
widely studied in information design literature

Proof Sketch via Two-Step LP

Proof ideas for optimal structure:

[three-interval **miscalibration structure**] $\Leftarrow$ mean-preserving contraction (MPC) constraint

Suppose structure is violated, we can construct another perfectly calibrated predictor $\hat{F}^*$ and miscalibration χ^* with desired miscalibration structure and

- objective is the same
- smaller calibration error

[three-interval **payoff structure**] $\Leftarrow$ MPC constraint + ECE budget constraint for χ

Proved by LP duality. The analysis shares similarity to auction design for budgeted buyers

- **monotone** $U'(q) \approx$ **monotone allocation rule** in auction design
- **linear tail** $U'(q) = -\alpha$ for $q \leq q_L$, $U'(q) = \alpha$ for $q \geq q_H$ α : dual variable for ECE budget constraint

Computing (Near)-Optimal Predictor

Theorem [Feng Tang, SODA'26] There exists LP-based algorithm for computing optimal ε -calibrated predictor with running time $\text{poly}(|\mathcal{X}|, |\mathcal{A}|)$.

$\mathcal{X}$: feature space, $\mathcal{A}$: action space

- Two-step LP + a novel two-layer discretization $\Rightarrow$ FPTAS

apply to more general ℓ_p -ECE constraints

- **Observation:** when $\varepsilon = 0$, **persuasive calibration** $\equiv$ **Bayesian persuasion (BP)**

OPT can be efficiently computed by applying **revelation principle** and then consider LP of **incentive compatible (IC)** action recommendation

- *Proof idea:* when $\varepsilon > 0$, persuasive calibration can be interpreted

a new variant of BP: **persuasion with signal-dependent bias**

“aggregate IC violation can be at most ε ”

Summary

- An intrinsic connection between **calibration** and **information design**
 - Calibrated predictor $\equiv$ signaling scheme where signal = posterior mean
 - [Two-step view] General predictor $\equiv$ calibrated predictor + miscalibration plan
- Applications:
 - Persuasive calibration: how to **design** predictors under incentive misalignment
 - How to **compare** different predictors (next part)
 - How to **design** predictors in digital advertising auction (next part)

Thanks!

Questions?

Please send me an email for any questions/comments:

ydfeng@ust.hk

wtang2359@gmail.com